INTERNATIONAL MONETARY FUND

KYRGYZ REPUBLIC

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STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by staff of the International Monetary Fund and the International Development Association

Risk of external debt distress	Moderate
Overall risk of debt distress	Moderate
Granularity in the risk rating	Some Space
Application of judgment	No

This debt sustainability analysis (DSA) indicates that the Kyrgyz Republic's public debt remains sustainable and at moderate risk of debt distress for both external public debt and total public debt. The external and total public debt outlooks remain vulnerable to export shocks and real GDP growth shocks, respectively. The breaches of the debt thresholds under these standard shocks imply moderate risk ratings for external and total public debt. The Kyrgyz Republic's current debt-carrying capacity is assessed as strong, and there remains some space to absorb shocks, despite the COVID crisis-induced spike in total public debt to 68 percent of GDP in 2020.[1] *Without consolidation measures public debt would continue to rise in the longer term. Strengthening debt management, avoiding non-concessional borrowing and improving public investment management would help reduce fiscal and external imbalances, thus containing debt vulnerabilities.*

[1] The DSA follows the IMF and World Bank Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework (DSF) for Low-Income Countries (LICs). See IMF, 2018, Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-Income Countries. The Kyrgyz Republic's debt-carrying capacity remains strong as its Composite Indicator is 3.123, which is based on the 2020 October WEO and the 2019 CPIA that was released in July 2020.

PUBLIC DEBT COVERAGE

1. The debt covers state government debt (both central and local government), state guarantees, and the debt of the central bank to the IMF (Text Table 1). Almost all public sector debt is issued by the central government. Local governments have no external debt and insignificant domestic debt. The social security fund has no debt. State-owned enterprises (SOEs) mostly borrow from the government, have no external debt and limited short-term domestic borrowing from the banking sector. In addition, the government has no outstanding guarantees, as the budget code prevents the state from guaranteeing debt of SOEs and other public entities since 2007, except when stipulated by the obligations of memberships in international and inter-governmental organizations. Nevertheless, a contingent liability shock of 7 percent of GDP was applied, reflecting risks stemming from the operation of SOEs (2 percent of GDP, which is about the structural cash shortfall of loss-making energy sector SOEs)[2] and the default value representing the average cost to the government during a financial crisis (5 percent of GDP, Text Table 2).

Text Table 1. Kyrgyz Republic: Public Debt Coverage

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	

Text Table 2. Kyrgyz Republic: Combined Contingent Liability Shock

1 The country's coverage of public debt	The central, state, and local governments, central bank, government-guaranteed debt			
	Default		Used for the analysis	Reasons for deviations from the default settings
2 Other elements of the general government not captured in 1.	0	percent of GDP	0.0	
3 SoE's debt (guaranteed and not guaranteed by the government) 1/	2	percent of GDP	2.0	default value (2 percent of GDP) reflects possible losses from SOE operations.
4 PPP	35	percent of PPP stock	0.0	
5 Financial market (the default value of 5 percent of GDP is the minimum value)	5	percent of GDP	5.0	
Total (2+3+4+5) (in percent of GDP)			7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

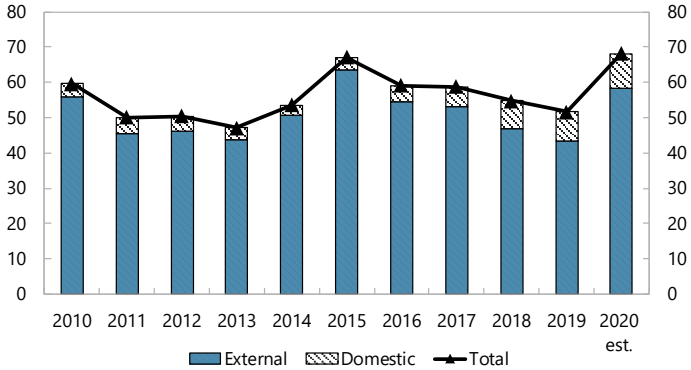
[2] IMF Country Report No. 19/208, Kyrgyz Republic—Staff Report for 2019 Article IV Consultation, ¶34.

BACKGROUND

2. After a steady decline since 2015, public debt increased significantly in 2020 (Text Chart 1). A lower output (GDP expressed in US$ declined by 13 percent), the 19 percent depreciation of Kyrgyz Som against the US dollar (year-on-year at end December), and an increase in the general government budget deficit (estimated at 3.3 percent of GDP) in 2020 resulted in a sharp increase in public debt to 68 (16.4 percentage points higher than in 2019). Domestic debt has increased from 3.6 percent in 2015 to 9.8 percent of GDP in 2020, accounting for about 14.5 percent of total debt. Domestic public debt is mostly held by commercial banks (50 percent) and the social security fund (30 percent).

Text Chart 1. Public Debt, (2010-2020)

Percent of GDP



Sources: Country authorities

UNDERLYING ASSUMPTIONS

3. The macroeconomic outlook underlying this DSA has weakened in the short term due to the COVID-crisis compared to the previous DSA. The revised projections assume that that the global pandemic recedes this year and activity rebounds in 2021-22 (Text Table 3):

Text Table 3. Kyrgyz Republic: Selected Indicators, 2019-2039

(percent of GDP)

	2019	2020	2021	2022	2021-26	2027-31	2032-41
Real GDP growth (percent)							
Current DSA	4.6	-8.6	3.8	6.4	4.4	4.0	4.0
Previous DSA[1]	4.5	0.4	6.0	4.3	n.a.	n.a.	n.a.
Overall fiscal balance (percent of GDP)[2]							
Current DSA	-0.1	-3.3	-4.2	-4.0	-3.6	-3.4	-3.3
Previous DSA[1]	-0.1	-7.8	-4.8	-3.0	n.a.	n.a.	n.a.
Current account balance (percent of GDP)							
Current DSA	-12.1	4.5	-5.8	-6.2	-6.2	-7.6	-4.7
Previous DSA[1]	-9.2	-14.5	-10.0	-7.5	n.a.	n.a.	n.a.
PIP Disbursements (millions of US$)							
Current DSA	188	162	296	271	240	250	359
Previous DSA[1]	268	362	415	395	n.a.	n.a.	n.a.

Sources: Kyrgyz authorities and IMF staff estimates.

1/ IMF Country Report No. 20/90, Kyrgyz Republic—Request for Purchase under the Rapid Financing Instrument and Disbursement under the Rapid Credit Facility Staff Report--Debt Sustainability Analysis.

2/ Including onlending to energy SOEs.

- **Growth and inflation.** Growth in 2020 was substantially weaker than in the previous DSA but is expected to rebound to 3.8 percent in 2021 and further increase to 6.4 percent in 2022, underpinned by a more favorable global outlook, a recovery of domestic activity as borders reopen, higher gold production, an increase in remittances from oil exporting neighbors, and a rebound in tourism, transportation and related services. Growth should gradually converge to its potential of 4 percent in the medium term and is expected to remain at its potential over the longer term in the absence of structural reforms. End-of-period inflation reached 9.7 percent in 2020 but is projected to decline to 7.4 percent in 2021 and remain in the authorities' target range (5 to 7 percent) in the medium and long term.

- **Fiscal policy.** The overall budget deficit increased from near balance in 2019 to 3.3 percent of GDP in 2020, and is projected to reach 4.2 percent of GDP in 2021, before gradually declining to about 3.3 percent of GDP by 2026 thanks to a gradual decrease in the size of the wage bill as a percentage of GDP, and a decrease in foreign-financed public investment. The overall deficit is projected to remain at the same level over the long term.

- **External sector.** The current account deficit swung from a deficit of 12 percent of GDP in 2019 to a surplus of 4.5 percent of GDP in 2020, primarily because of significant import compression, which reflects the depreciation of the exchange rate, border closures during the pandemic, and reduced demand. Imports are projected to rebound in 2021 and the current account deficit should reach upward of 6 percent of GDP in the medium term (2026).

- **Financing assumptions.** The new external borrowing is assumed to remain mostly on semi-concessional terms and the country is expected to remain IDA-eligible over the projection horizon, but net external financing is expected to decline. As a result, the share of domestic borrowing is expected to increase from 26 percent of the total in 2021 to about 70 percent in 2026. If financing becomes a constraint, the deficit is likely to be lower and therefore domestic debt as well. Domestic borrowing is assumed at interest rates of 6 to 11 percent for maturities ranging between 1 year and 10 years in the medium term, with a decrease thereafter for bonds ranging from 100 to 200 basis points as the financial market develops.

4. Realism tools suggest that the baseline projections are reasonable:

- *Drivers of debt dynamics (Figure 3).* The realism tool on the drivers of debt dynamics does not raise any flag. The forecast error on the change in the ratio of public debt-to-GDP over the past five years is small (even negative for the external public debt ratio). The projected public debt ratio is expected to decline more significantly (-4 percentage points of GDP) than previously evidenced in a more significative way (+1 percentage points of GDP), but the difference is small and well substantiated by the expected recovery of economic growth.

- *Realism of planned fiscal adjustment* (Figure 4). The projected evolution of the primary balance shows no adjustment and is near the median of the distribution of the past adjustments to primary fiscal deficit of the sample of LICs, and thus does not raise concerns.

- *Consistency between fiscal adjustment and growth* (Figure 4). The growth projection for 2021 is well above the growth path suggested by different fiscal multipliers as post-crisis growth is projected to rebound to more than 6 percent in 2022, driven by tourism, transport and industry. The result generated by this realism tool is not reliable, as the impact of the COVID-19 pandemic is not well captured by this exercise.

- *Consistency between public investment and growth* (Figure 4). The contribution of public investment to growth is slightly higher than the previous DSA, but still below historical contribution.

COUNTRY CLASSIFICATION AND STRESS TESTS

5. The Kyrgyz Republic's debt-carrying capacity is assessed as strong (Text Table 4). The country's Composite Indicator (CI) index[3] is 3.12, above the threshold of 3.05 for strong debt-carrying capacity. The CI is calculated for the last two IMF World Economic Outlook (WEO) vintages (October 2020 and April 2020) and the World Bank's 2019 CPIA. This translates into the following external debt burden thresholds: 240 percent of the present value (PV) of external debt-to-exports ratio, 55 percent of the PV of external debt-to-GDP, 21 percent of the PV of external debt service-to-exports, and 23 percent of the PV of debt service-to-revenue. The total public debt burden threshold is 70 percent of the PV of total public debt-to-GDP ratio.

Text Table 4. Kyrgyz Republic: Debt-Carrying Capacity and Relevant Indicative Thresholds

Debt Carrying Capacity	Strong		
Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Strong	Strong 3.12	Strong 3.19	Strong 3.19

EXTERNAL debt burden thresholds	Strong
PV of debt in % of	
Exports	240
GDP	55
Debt service in % of	
Exports	21
Revenue	23

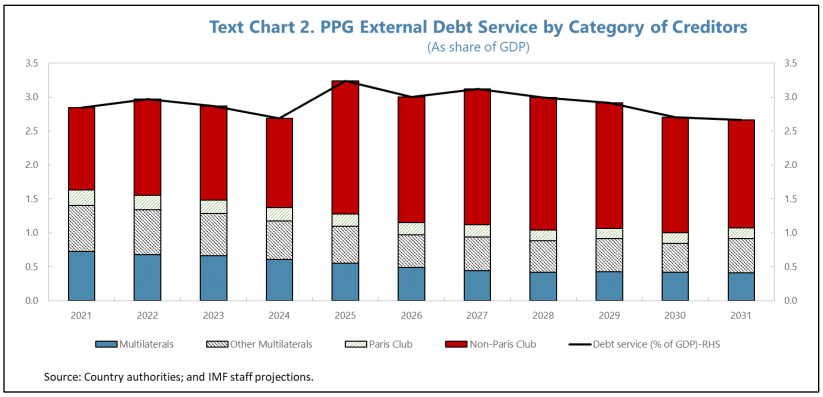
TOTAL public debt benchmark	Strong
PV of total public debt in percent of GDP	70

[3] The CI is a function of the World Bank's Country Policy and Institutional Assessment (CPIA) score, international reserves, remittances, country and global economic growth. The calculation is based on 10-year averages of the variables, across 5 years of historical data and 5 years of projection. For more details, see IMF, 2018, Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-Income Countries.

EXTERNAL DSA

6. All four external debt ratios remain well under their respective thresholds under the baseline scenario (Figure 1, and Tables 1 and 3). External Public and Publicly Guaranteed (PPG) debt was 58 percent of GDP at end-2020 and another 29.5 percentage points are private external debt. Private external debt is expected to gradually decline to under 22 percent of GDP by 2026 as net external private borrowing is projected to grow less than nominal GDP growth under the baseline scenario. While the PV of PPG external debt increased to 36 percent of GDP in 2020 (from about 32 percent at end-2019 estimated in the previous DSA), it is also projected to decline to about 28 of percent of GDP by end-2026, well below the threshold of 55 percent of GDP. This is consistent with the gradual decrease under the baseline of the ratio of nominal debt to GDP after an uptick in debt service in 2025 due to higher payments due to non-Paris Club creditors (Text Chart 2).

7. While external debt stock and debt service ratios remain under their respective thresholds in the baseline scenario, two ratios breach their respective thresholds for continuous years and significantly under the most impactful standard stress test ("most extreme shock"): a shock to exports. Under this shock, the applicable thresholds for the Kyrgyz Republic are breached significantly for the PV of debt-to-exports and the debt service-to-exports, the latter of which rises above the 21 percent benchmark to 25 percent in 2029, and declines thereafter. The PV of debt-to-exports breaches its threshold during 2023-2028 by an average deviation of 10 percent, and subsequently gradually decreases. The PV of PPG external debt-to-GDP ratio also breaches its threshold, but for only one year and by a marginal amount (55.7 percent vs. a threshold value of 55 percent). The risks posed by private external debt decrease, given diminishing outstanding levels over time.



PUBLIC DSA

8. The two debt stock ratios used to assess the risk of total public debt distress remain stable under the baseline scenario, but the debt service ratio increases steadily (Figure 2 and Tables 2 and 4). In 2020 public debt increased by nearly 17 percentage points to 68 percent of GDP due to output contraction during the COVID-19 crisis, the depreciation of the exchange rate and the

Text Table 5. Kyrgyz Republic: Comparison of Debt Ratio
(In percent of GDP)

	2020	2021	2022	2023	2024	2025	2026	Long Term (2031) 2/
PPG external debt-to-GDP ratio								
Current DSA	58.3	56.8	52.6	49.4	46.9	44.7	42.8	34.6
Previous DSA[1]	56.5	53.5	51.8	50.5	48.9	47.6	42.6	42.0
Public debt-to-GDP ratio								
Current DSA	68.0	67.1	65.3	64.4	64.6	64.6	65.0	69.1
Previous DSA[1]	65.9	64.8	64.0	62.9	61.9	60.9	60.3	59.6

Sources: Kyrgyz authorities and IMF staff estimates.
1/ IMF Country Report No. 20/90, Kyrgyz Republic—Request for Purchase under the Rapid Financing Instrument and Disbursement under the Rapid Credit Facility Staff Report--Debt Sustainability Analysis.
2/ for the previous DSA: 2030.

increase in the primary fiscal deficit (Text Table 5). It is expected to remain around 65 percent of GDP over the medium term, but rise to 77 percent of GDP by 2041. The PV of total public debt is projected to stay under 56 percent of GDP target of the debt strategy through 2026 and below the sustainability threshold of 70 percent of GDP. Moreover, debt service should remain manageable in the medium term: the ratio of debt service-to-revenue (including grants) averages 24 percent in the next five years. It is projected to rise in the medium term, reflecting an increase in gross financing need due to more expensive domestic financing compared to external financing.

9. Total public debt is particularly vulnerable to a growth shock and the PV of total public debt over GDP breaches its benchmark within the stress test horizon. The most severe test is a simulated growth shock. Under this shock, the PV of debt-to-GDP ratio reaches 70 percent of GDP by 2023 and is on a persistent upward trajectory. Moreover, such a shock would put the two other indicators on an upward path: the PV of debt-to-revenue ratio would reach 420 in 2031, and the debt service-to-revenue ratio raises above 50 percent by 2028. Although no explicit benchmark exists for these two ratios, the projections point to potentially severe debt and liquidity difficulties in the long run under a GDP shock.

A. Risk Rating and Vulnerabilities

10. The DSA indicates that external debt remains at a moderate risk of debt distress. The breaches of their respective thresholds by the PV of debt-to-exports ratio and the debt service-to-export ratio under the most impactful standard stress test cause the assessment of the risk of external debt distress to be moderate. Conversely, the PV of external debt-to-GDP ratio is assessed

to be resilient even under its "most extreme" shock, because this ratio breaches its threshold for only one year (i.e., 2023) and marginally (i.e., 0.7 percentage point of GDP).

11. External debt is deemed to be sustainable. The profiles of all ratios under the baseline are projected to show a downward trajectory before 2031, driven by the decline in PPG external debt.

12. Overall public debt remains at a moderate risk of debt distress primarily because of the moderate risk of external debt distress. However, total public debt is also vulnerable to a real GDP growth shock; the PV of debt-to-GDP ratio breaches its threshold of 70 percent by an increasing margin under such shock, confirming the determination of a moderate risk of total debt distress. The steady increase in the debt service ratio over time under the baseline scenario further supports this determination. However, the resilience of the public debt risk assessment to other standardized stress tests indicates that public debt would remain sustainable.

13. A granular assessment of the moderate risk rating shows that the Kyrgyz Republic has some space to absorb shocks (Figure 5). The external PPG debt outlook is resilient under the baseline. While it is vulnerable to large external shocks, especially to exports, the gap between debt burden indicators and their respective thresholds indicates that the Kyrgyz Republic has some space to absorb shocks without being downgraded to high risk of debt distress.

14. The authorities need to maintain fiscal discipline, strengthen public debt management, and improve the business climate. The heightened uncertainty in the global economic outlook affects prospects for economic recovery and exports, both of which are critical elements of Kyrgyzstan's debt sustainability. To keep public debt sustainable, the deficit should be reduced under 3.5 percent of GDP in the medium term after the economy recovers from the COVID-19 crisis. To meet the country's growing spending needs on health and education, infrastructure, and social assistance without undermining debt sustainability, further efforts are needed to strengthen public debt management, while keeping new borrowing on concessional terms; improve public investment management to contain contingent liabilities and spur growth; raise spending efficiency; and create additional fiscal space by expenditure prioritization, as well as tax policy and administrative measures to raise more tax revenue. These efforts would strengthen engagement with donors and help mobilize additional concessional financing. Equally important are structural reforms to improve the business environment and strengthen the competitiveness of Kyrgyz exports over the medium and long term.

AUTHORITIES' VIEWS

15. The authorities broadly shared the views of Bank and Fund staff. They expressed a strong commitment to debt sustainability and prudent debt management. To guide fiscal policy, the authorities intend to adopt a fiscal rule that will put a ceiling on the budget deficit. They also intend to set a legally binding overall debt ceiling for nominal debt as a percentage of GDP.

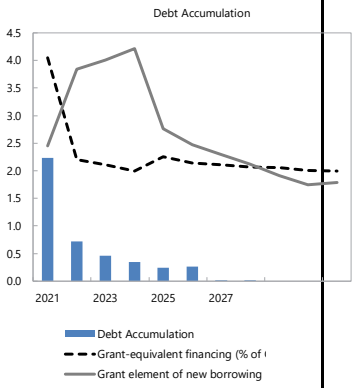
Table 1. Kyrgyz Republic: External Debt Sustainability Framework, Baseline Scenario, 2018-2041

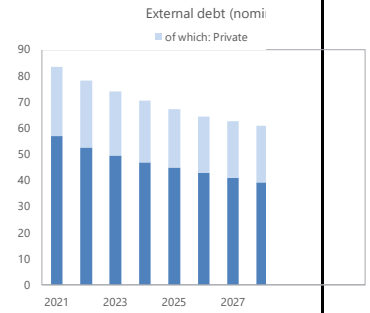
(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**76.5**	**72.8**	**87.8**	**83.3**	**78.1**	**73.9**	**70.4**	**67.2**	**64.3**	**56.1**	**55.7**	**81.0**	**66.7**
of which: public and publicly guaranteed (PPG)	**47.0**	**43.3**	**58.3**	**56.8**	**52.6**	**49.4**	**46.9**	**44.7**	**42.8**	**34.6**	**29.5**	**50.6**	**43.8**
Change in external debt	-4.6	-3.7	14.9	-4.4	-5.2	-4.2	-3.5	-3.2	-2.9	-1.7	3.6		
Identified net debt-creating flows	**4.8**	**3.1**	**12.7**	**1.9**	**-1.1**	**0.2**	**0.4**	**0.6**	**2.0**	**2.0**	**0.5**	**2.3**	**2.1**
Non-interest current account deficit	**11.4**	**11.5**	**-5.4**	**5.1**	**5.6**	**5.5**	**5.5**	**5.6**	**6.4**	**5.8**	**3.3**	**10.1**	**6.3**
Deficit in balance of goods and services	38.3	28.9	20.2	31.6	29.8	28.5	27.7	27.1	27.7	27.2	23.9	**34.2**	**28.8**
Exports	33.2	35.2	31.4	32.0	36.9	36.4	36.0	35.8	34.7	34.3	41.1		
Imports	71.5	64.1	51.6	63.7	66.7	64.9	63.8	62.9	62.4	61.5	65.1		
Net current transfers (negative = inflow)	-29.3	-25.6	-29.0	-29.4	-27.3	-26.1	-25.3	-24.6	-24.1	-23.5	-22.2	**-28.7**	**-25.1**
of which: official	-0.6	-1.3	-1.0	-0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	2.4	8.2	3.4	2.9	3.0	3.0	3.1	3.1	2.8	2.1	1.6	**4.7**	**2.5**
Net FDI (negative = inflow)	**-1.7**	**-3.8**	**6.6**	**-0.6**	**-2.4**	**-2.7**	**-2.9**	**-2.9**	**-2.6**	**-1.6**	**-1.2**	**-4.9**	**-2.0**
Endogenous debt dynamics 2/	**-4.9**	**-4.6**	**11.5**	**-2.6**	**-4.3**	**-2.5**	**-2.2**	**-2.0**	**-1.8**	**-2.2**	**-1.6**		
Contribution from nominal interest rate	0.6	0.6	0.9	0.7	0.7	0.6	0.6	0.6	0.6	0.5	0.4		
Contribution from real GDP growth	-2.6	-3.3	7.2	-3.3	-4.9	-3.2	-2.8	-2.6	-2.4	-2.6	-2.0		
Contribution from price and exchange rate changes	-3.0	-1.9	3.4	...	...	...	...	...	...	...	...		
Residual 3/	**-9.4**	**-6.8**	**2.3**	**-6.4**	**-4.1**	**-4.4**	**-3.9**	**-3.8**	**-4.9**	**-3.8**	**3.1**	**-2.2**	**-4.9**
of which: exceptional financing	-3.0	0.1	-0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**36.0**	**36.9**	**34.6**	**32.6**	**31.1**	**29.6**	**28.2**	**22.2**	**19.4**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**114.6**	**115.1**	**93.8**	**89.7**	**86.4**	**82.6**	**81.3**	**64.7**	**47.1**		
PPG debt service-to-exports ratio	**15.0**	**5.7**	**9.2**	**8.9**	**8.0**	**7.9**	**7.5**	**9.0**	**8.6**	**7.9**	**4.1**		
PPG debt service-to-revenue ratio	**16.2**	**6.6**	**9.9**	**9.8**	**9.9**	**9.6**	**9.0**	**10.9**	**10.1**	**9.3**	**5.8**		
Gross external financing need (Million of U.S. dollars)	1548.0	1367.3	700.0	950.8	839.8	826.6	851.4	1022.2	1186.9	1538.5	852.7		
Key macroeconomic assumptions													
Real GDP growth (in percent)	3.8	4.6	-8.6	3.8	6.4	4.4	4.0	4.0	3.7	4.8	4.0	**3.4**	**4.2**
GDP deflator in US dollar terms (change in percent)	3.8	2.5	-4.4	-0.2	2.1	3.0	1.9	2.0	1.9	1.0	-1.0	**2.0**	**1.4**
Effective interest rate (percent) 4/	0.9	0.9	1.1	0.8	0.9	0.9	0.9	0.9	0.9	0.9	0.8	**0.8**	**0.9**
Growth of exports of G&S (US dollar terms, in percent)	4.1	13.8	-22.2	5.7	25.0	6.0	5.0	5.4	2.5	13.8	7.4	**1.8**	**6.8**
Growth of imports of G&S (US dollar terms, in percent)	15.6	-3.8	-29.7	27.8	13.8	4.6	4.1	4.6	4.9	5.7	4.9	**2.4**	**7.6**
Grant element of new public sector borrowing (in percent)	...	...	...	36.2	37.4	37.6	37.7	36.5	36.2	35.6	34.4	**...**	**36.4**
Government revenues (excluding grants, in percent of GDP)	30.8	30.3	29.0	28.9	30.0	29.8	29.7	29.7	29.6	29.0	29.0	**31.2**	**29.3**
Aid flows (in Million of US dollars) 5/	137.4	190.1	154.3	306.7	228.0	228.9	225.5	267.3	262.6	298.7	404.4		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	4.0	2.2	2.1	2.0	2.3	2.1	2.0	1.8	**...**	**2.3**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	55.4	53.4	55.6	57.6	53.0	53.7	55.4	57.2	**...**	**54.7**
Nominal GDP (Million of US dollars)	8,271	8,872	7,747	8,027	8,723	9,376	9,934	10,531	11,129	14,247	22,359		
Nominal dollar GDP growth	7.4	7.3	-12.7	3.6	8.7	7.5	5.9	6.0	5.7	5.8	3.0	**5.5**	**5.7**
Memorandum items:													
PV of external debt 7/	...	...	65.5	63.4	60.1	57.1	54.6	52.1	49.8	43.7	45.6		
In percent of exports	...	...	208.6	197.9	163.1	157.2	151.7	145.5	143.2	127.5	110.7		
Total external debt service-to-exports ratio	27.1	21.9	24.9	22.8	17.5	16.6	16.7	19.8	19.8	19.3	4.1		
PV of PPG external debt (in Million of US dollars)			2787.6	2960.5	3018.2	3058.4	3090.8	3114.9	3142.8	3161.1	4333.8		
(PVt-PVt-1)/GDPt-1 (in percent)				2.2	0.7	0.5	0.3	0.2	0.3	0.0	0.6		
Non-interest current account deficit that stabilizes debt ratio	16.0	15.2	-20.3	9.6	10.8	9.7	9.0	8.7	9.3	7.5	-0.4		

Definition of external/domestic debt

Is there a material difference between the two criteria?





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + Ɛα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, Ɛ=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 2. Kyrgyz Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018-2041

(in percent of GDP, unless otherwise indicated)

	Actual			Projections							
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041
Public sector debt 1/	**54.8**	**51.6**	**68.0**	**67.1**	**65.1**	**63.5**	**63.6**	**64.3**	**64.8**	**66.8**	**63.5**
of which: external debt	**47.0**	**43.3**	**58.3**	**56.8**	**52.6**	**49.4**	**46.9**	**44.7**	**42.8**	**34.6**	**29.5**
Change in public sector debt	-4.0	-3.2	16.4	-1.0	-2.0	-1.6	0.1	0.7	0.5	-0.5	-1.1
Identified debt-creating flows	**-5.7**	**-4.7**	**13.4**	**-1.1**	**-1.3**	**-0.4**	**0.0**	**0.0**	**0.2**	**0.1**	**-0.4**
Primary deficit	**-0.4**	**-0.7**	**2.3**	**3.3**	**3.0**	**2.4**	**2.3**	**2.1**	**2.0**	**1.7**	**0.3**
Revenue and grants	32.5	32.4	31.0	31.1	31.0	30.9	30.8	30.8	30.7	30.1	30.1
of which: grants	1.7	2.1	2.0	2.2	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Primary (noninterest) expenditure	32.0	31.7	33.3	34.5	34.1	33.2	33.1	32.9	32.7	31.8	30.4
Automatic debt dynamics	**-2.6**	**-3.9**	**11.1**	**-4.4**	**-4.3**	**-2.8**	**-2.2**	**-2.1**	**-1.9**	**-1.6**	**-0.7**
Contribution from interest rate/growth differential	-3.7	-1.2	4.1	-4.4	-4.3	-2.8	-2.2	-2.1	-1.9	-1.6	-0.7
of which: contribution from average real interest rate	-1.8	1.2	-0.8	-1.9	-0.2	-0.1	0.2	0.3	0.4	1.5	1.8
of which: contribution from real GDP growth	-2.0	-2.4	4.9	-2.5	-4.1	-2.7	-2.4	-2.4	-2.3	-3.1	-2.5
Contribution from real exchange rate depreciation	1.1	-2.7	7.0	...	...	...	...	...	...	...	...
Other identified debt-creating flows	**-2.7**	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.2	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	-2.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual	**1.8**	**1.5**	**3.0**	**0.1**	**-0.7**	**-1.1**	**0.0**	**0.6**	**0.3**	**-0.6**	**-0.7**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**...**	**...**	**48.3**	**48.0**	**47.6**	**47.2**	**48.3**	**49.6**	**50.6**	**54.8**	**53.9**
PV of public debt-to-revenue and grants ratio	**...**	**...**	**155.7**	**154.3**	**153.5**	**153.0**	**156.7**	**160.8**	**164.9**	**181.9**	**178.9**
Debt service-to-revenue and grants ratio 3/	**15.4**	**6.2**	**9.3**	**11.6**	**15.2**	**16.9**	**17.8**	**21.1**	**23.3**	**33.9**	**36.5**
Gross financing need 4/	1.9	1.2	5.2	7.0	7.7	7.6	7.8	8.6	9.2	11.9	11.3
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	3.8	4.6	-8.6	3.8	6.4	4.4	4.0	4.0	3.7	4.8	4.0
Average nominal interest rate on external debt (in percent)	1.3	1.5	1.8	1.2	1.3	1.3	1.3	1.3	1.4	1.4	1.5
Average real interest rate on domestic debt (in percent)	-3.6	-3.8	-5.4	-1.6	2.0	2.4	3.9	4.0	4.3	5.7	5.8
Real exchange rate depreciation (in percent, + indicates depreciation)	2.5	-5.8	14.9	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.4	3.9	5.8	9.4	6.2	6.1	5.0	5.0	5.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-8.3	3.5	-4.0	7.5	5.2	1.9	3.6	3.4	3.1	4.2	3.4
Primary deficit that stabilizes the debt-to-GDP ratio 5/	3.5	2.5	-14.2	4.3	5.0	3.9	2.2	1.5	1.5	2.2	1.4
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments, central bank, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

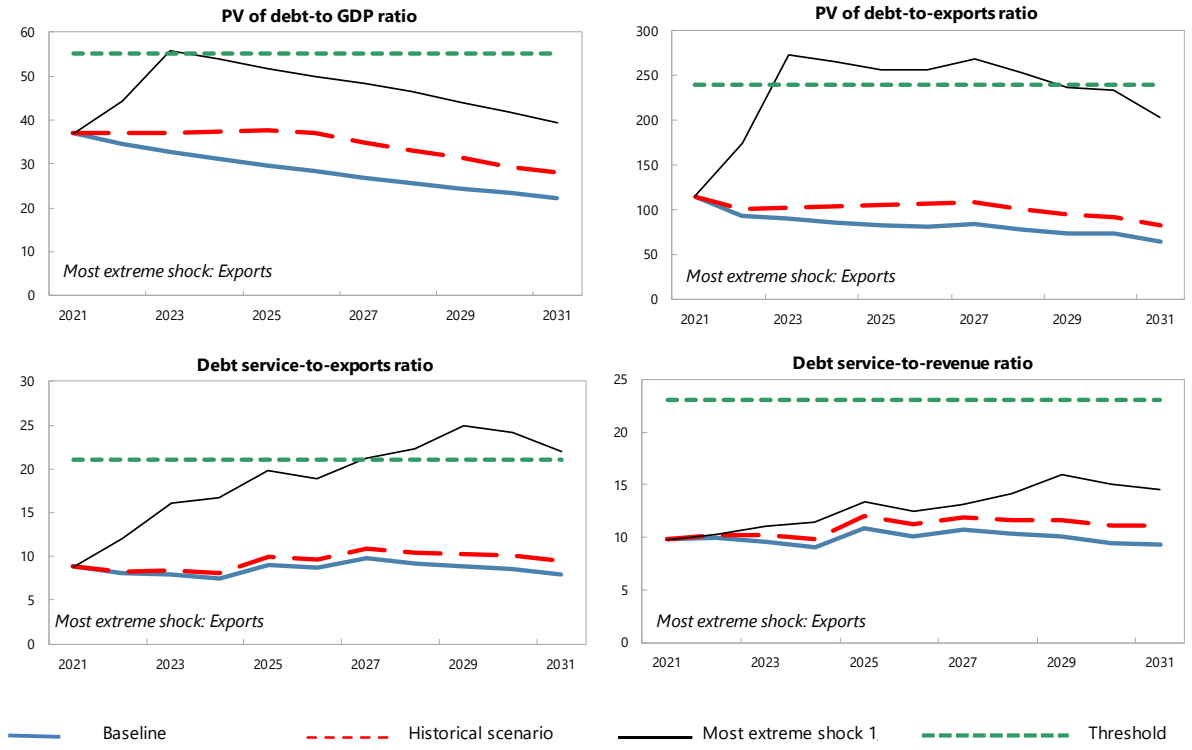
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2021–2031



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	n.a.	n.a.
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

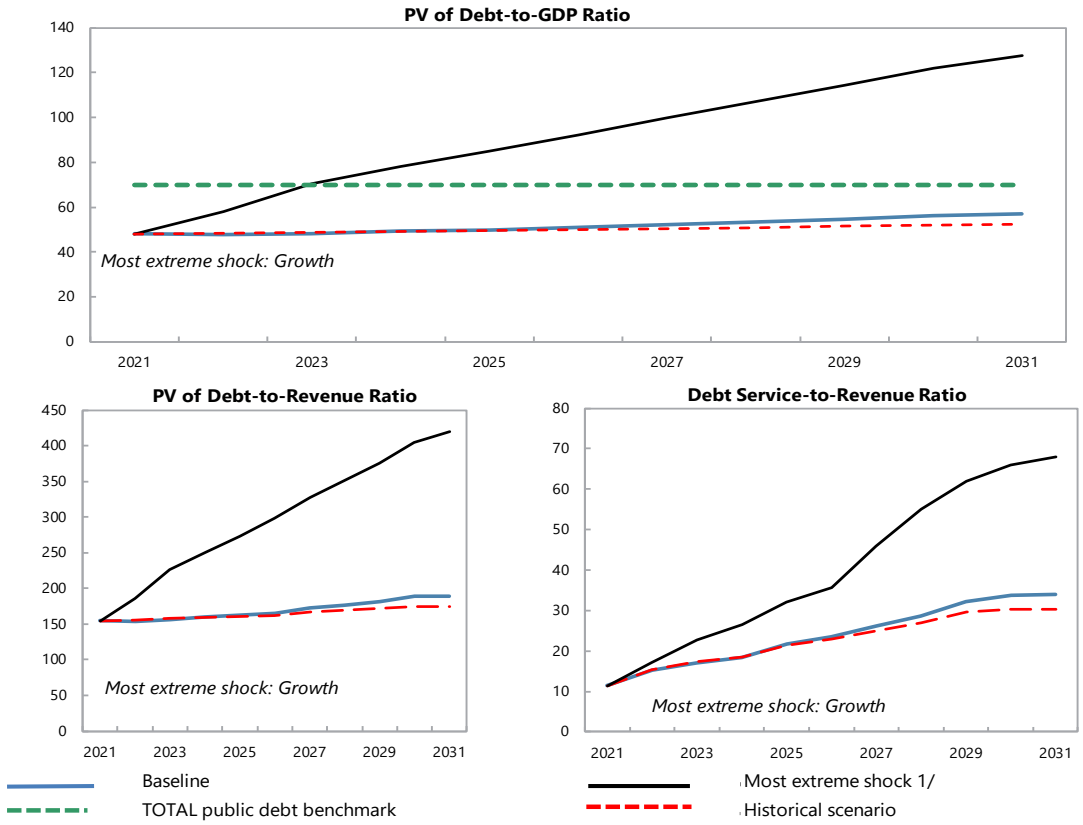
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	27	27
Avg. grace period	5	5

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Kyrgyz Republic: Indicators of Public Debt Under Alternative Scenarios, 2021-2031



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	33%	33%
Domestic medium and long-term	60%	60%
Domestic short-term	7%	7%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.7%	1.7%
Avg. maturity (incl. grace period)	27	27
Avg. grace period	5	5
Domestic MLT debt		
Avg. real interest rate on new borrowing	4.0%	4.0%
Avg. maturity (incl. grace period)	5	5
Avg. grace period	4	4
Domestic short-term debt		
Avg. real interest rate	0.8%	0.8%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–2031

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	37	35	33	31	30	28	27	26	24	23	22
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	37	37	37	37	38	37	35	33	31	29	28
B. Bound Tests											
B1. Real GDP growth	37	39	41	39	37	35	34	32	31	29	28
B2. Primary balance	37	35	33	32	31	29	28	27	26	25	24
B3. Exports	37	44	**56**	54	52	50	48	46	44	42	39
B4. Other flows 3/	37	40	43	42	40	39	37	36	34	32	30
B5. Depreciation	37	44	35	33	31	30	28	26	25	24	23
B6. Combination of B1-B5	37	51	53	51	49	48	46	44	41	39	37
C. Tailored Tests											
C1. Combined contingent liabilities	37	36	34	33	31	30	29	28	27	26	25
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	115	94	90	86	83	81	84	79	74	74	65
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	115	100	102	104	105	107	109	101	95	91	82
B. Bound Tests											
B1. Real GDP growth	115	94	90	86	83	81	84	79	74	74	65
B2. Primary balance	115	95	92	89	85	84	88	83	78	78	69
B3. Exports	115	174	**273**	**265**	**257**	**256**	**269**	**253**	237	233	204
B4. Other flows 3/	115	109	120	116	112	111	116	109	102	101	88
B5. Depreciation	115	94	76	73	69	68	69	64	61	61	54
B6. Combination of B1-B5	115	167	123	185	179	177	185	173	162	160	140
C. Tailored Tests											
C1. Combined contingent liabilities	115	98	94	91	87	86	91	86	82	82	72
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	9	8	8	7	9	9	10	9	9	9	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	9	8	8	8	10	10	11	10	10	10	9
B. Bound Tests											
B1. Real GDP growth	9	8	8	7	9	9	10	9	9	9	8
B2. Primary balance	9	8	8	8	9	9	10	9	9	9	8
B3. Exports	9	12	16	17	20	19	**21**	**22**	**25**	**24**	**22**
B4. Other flows 3/	9	8	8	8	10	9	10	11	11	11	10
B5. Depreciation	9	8	8	7	9	8	9	9	8	7	7
B6. Combination of B1-B5	9	11	13	13	15	15	16	18	18	17	16
C. Tailored Tests											
C1. Combined contingent liabilities	9	8	8	8	9	9	10	9	9	9	8
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	10	10	10	9	11	10	11	10	10	9	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	10	10	10	10	12	11	12	12	12	11	11
B. Bound Tests											
B1. Real GDP growth	10	11	12	11	14	13	13	13	13	12	12
B2. Primary balance	10	10	10	9	11	10	11	10	10	10	10
B3. Exports	10	10	11	11	13	13	13	14	16	15	15
B4. Other flows 3/	10	10	10	10	12	11	12	12	13	12	12
B5. Depreciation	10	12	12	11	13	12	13	13	11	10	10
B6. Combination of B1-B5	10	11	13	12	14	13	14	16	16	15	14
C. Tailored Tests											
C1. Combined contingent liabilities	10	10	10	9	11	10	11	11	10	10	9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2021–2031

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	48	48	48	49	50	51	52	53	55	56	57
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	48	48	49	49	49	50	50	51	51	52	52
B. Bound Tests											
B1. Real GDP growth	48	58	**70**	**78**	**85**	**92**	**100**	**107**	**114**	**122**	**128**
B2. Primary balance	48	50	52	53	53	54	55	56	58	59	60
B3. Exports	48	56	68	68	69	69	**70**	**71**	**71**	**72**	**71**
B4. Other flows 3/	48	53	59	60	61	61	63	63	64	65	65
B5. Depreciation	48	55	53	52	50	49	49	48	48	48	47
B6. Combination of B1-B5	48	49	51	53	55	57	59	61	63	65	67
C. Tailored Tests											
C1. Combined contingent liabilities	48	54	54	55	56	57	58	59	60	62	62
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	154	154	156	160	162	166	173	177	181	189	189
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	154	156	158	160	161	162	167	169	171	175	175
B. Bound Tests											
B1. Real GDP growth	154	186	226	251	274	298	328	351	375	404	420
B2. Primary balance	154	160	167	171	173	176	183	187	191	199	199
B3. Exports	154	181	219	222	223	226	233	235	236	241	237
B4. Other flows 3/	154	172	192	195	196	200	207	210	212	218	216
B5. Depreciation	154	179	172	169	164	161	162	160	159	160	156
B6. Combination of B1-B5	154	159	166	172	177	184	194	201	209	218	221
C. Tailored Tests											
C1. Combined contingent liabilities	154	174	175	179	181	184	191	195	199	206	207
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	11	15	17	18	22	23	26	29	32	34	34
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	11	15	17	18	22	23	25	27	30	30	30
B. Bound Tests											
B1. Real GDP growth	11	17	23	26	32	36	46	55	62	66	68
B2. Primary balance	11	15	18	20	23	24	29	32	34	35	35
B3. Exports	11	15	18	20	23	25	28	31	37	38	38
B4. Other flows 3/	11	15	17	19	23	24	27	30	35	36	36
B5. Depreciation	11	16	19	19	23	25	27	27	32	33	33
B6. Combination of B1-B5	11	15	18	19	23	25	29	31	37	39	40
C. Tailored Tests											
C1. Combined contingent liabilities	11	15	20	20	23	25	36	31	34	35	36
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

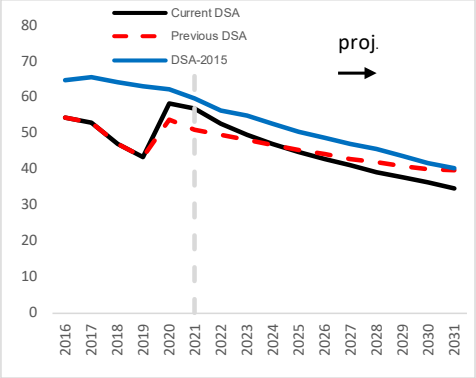
1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

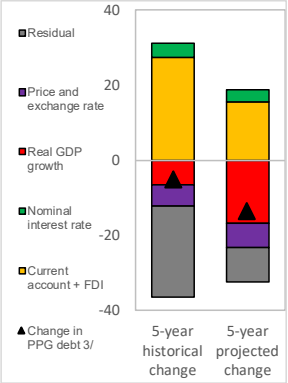
3/ Includes official and private transfers and FDI.

Figure 3. Kyrgyz Republic: Drivers of Debt Dynamics—Baseline Scenario

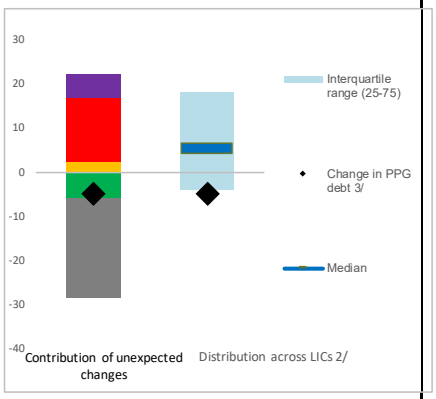
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

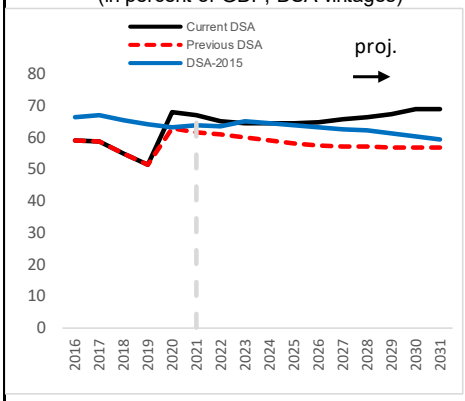


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

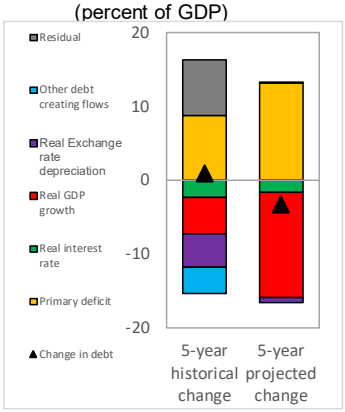


Public debt

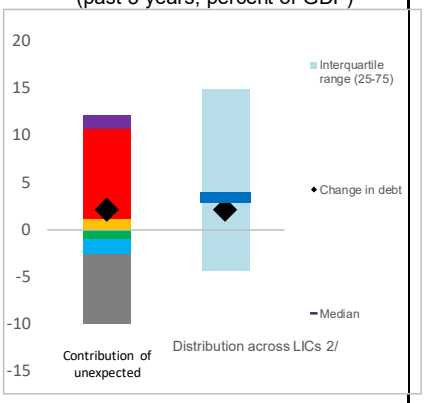
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



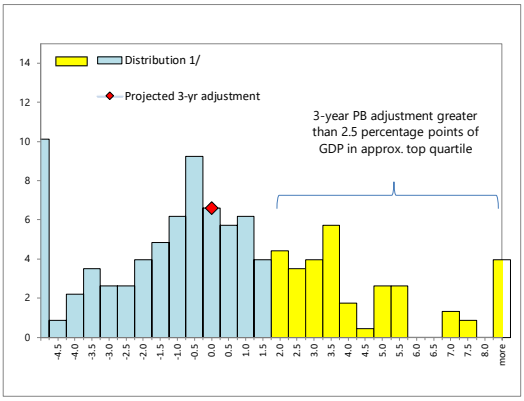
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

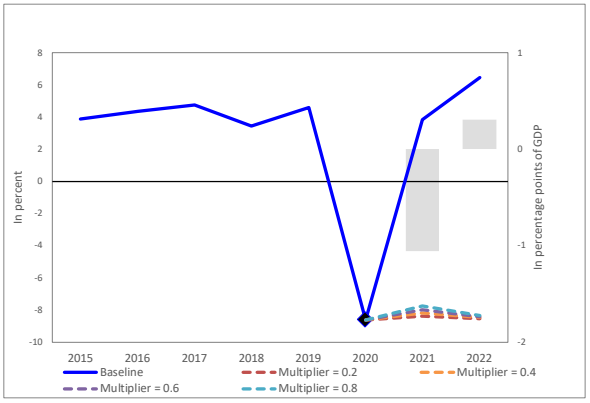
Figure 4. Kyrgyz Republic: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



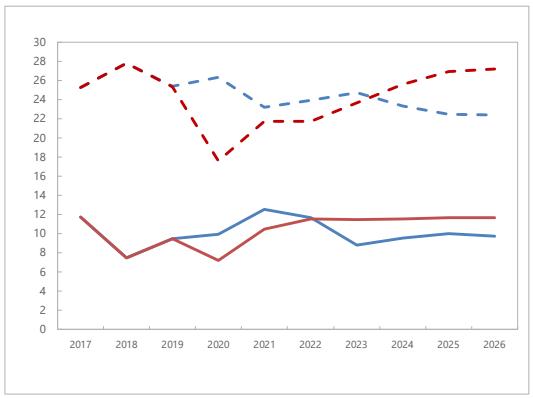
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

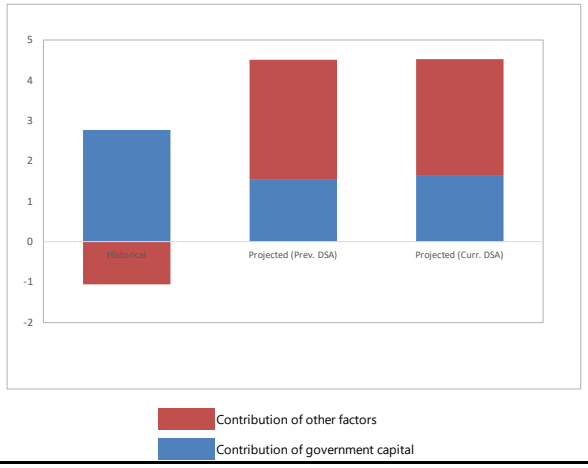
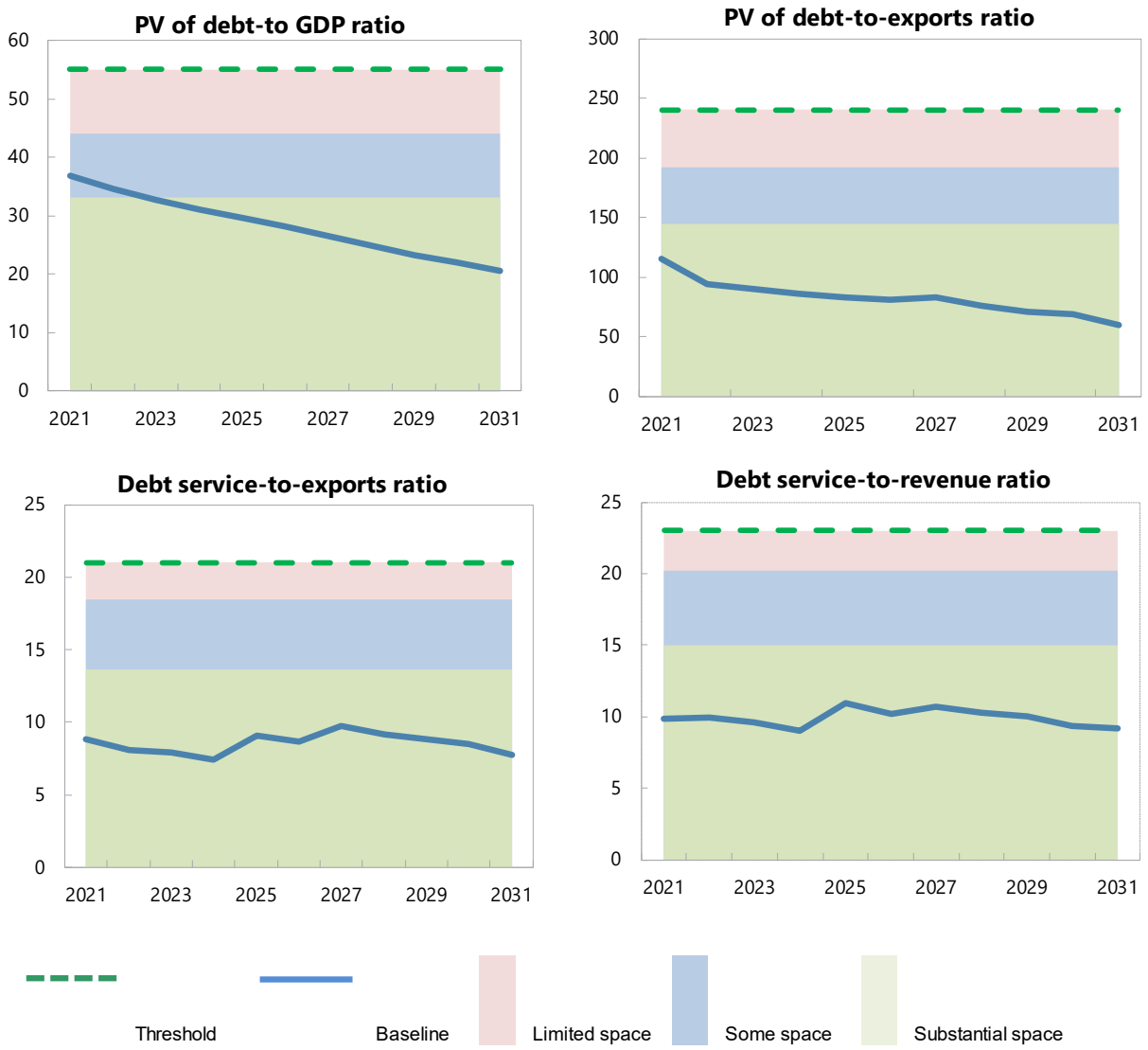


Figure 5. Kyrgyz Republic: Qualification of the Moderate Category, 2021–2031 1/



Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.